SHAILZA JOLLY

Berlin, Germany

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PROFILE

ML Scientist / AI Engineer, Ph.D. in Computer Science, 6+ years across industry research and product. Focused on making LLM systems accurate and auditable enough to trust: data-quality pipelines over billions of tokens at Amazon AI, LLM-powered product features for a medical-device platform at Flinn.ai, and Sift, an open-source hybrid retrieval system that answers only from retrieved passages and cites the source page. Depth in LLM internals and inference efficiency (KV-cache, speculative decoding, FlashAttention) and agentic system design. Publications at AAAI, NAACL, and EMNLP.

SKILLS

- **ML/GenAI:** LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation, model training and fine-tuning (T5, BERT)
- **LLM systems:** Agentic workflows (MCP, LangGraph/LangSmith), Retrieval-Augmented Generation (RAG), grounding and citations, persistent agent memory, experiment design, offline evaluation
- **Data & scale:** Python, PyTorch, PySpark; large-scale text processing/data quality.
- **Search & Retrieval:** Hybrid retrieval (BM25 + dense), cross-encoder reranking, vector search (Chroma, Pinecone), Sentence Transformers, Docling
- **Infrastructure & Deployment:** AWS, Docker, FastAPI, Git

WORK EXPERIENCE

Applied AI & LLM Systems — July 2025 - Present
Independent — *Berlin, Germany*

- Built and open-sourced Sift: grounded question-answering over technical PDF corpora. Docling chunking, hybrid BM25 and dense (Chroma) retrieval, cross-encoder reranking, served over MCP to an agent that answers only from retrieved passages, with a source and page citation on every claim.
- Built an agentic tutoring system in daily use: an LLM agent orchestrating retrieval over a structured knowledge base, persistent memory of errors & progress, adaptive lesson selection that targets weak areas, and voice I/O.
- Deepened modern LLM architecture and inference-efficiency internals through Stanford coursework: KV-cache, speculative decoding, FlashAttention.

Flinn.ai — January 2025 - June 2025
Senior AI Engineer — *Berlin, Germany*

- Led development of LLM-powered product features for a platform serving medical device manufacturers, including data extraction from medical research papers and multilingual complaint monitoring.
- Partnered with product and backend teams to scope problems, define success metrics, and deliver features end to end.
- Navigated cost-performance tradeoffs under production constraints.

Parental Leave (Maternity) — December 2023 - December 2024

- Career break for full-time childcare after the birth of my child.

Amazon AI — July 2022 - November 2023
Research Scientist — *Berlin, Germany*

- Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
- Presented findings and recommendations to senior science and engineering leadership; mentored Master's and Ph.D. interns on research deliverables.

German Research Center for Artificial Intelligence (DFKI) — March 2019 - June 2022
Machine Learning Scientist — *Berlin, Germany*

- Implemented transformer-based ML systems end-to-end (GPU training, Docker, production-level code) within BMBF-funded projects; published results at AAAI, NAACL, and EMNLP.
- Collaborated with academic and industry partners internationally; supervised interns and BSc/MSc students.

NVIDIA Research May - August 2021
Machine Learning Scientist Intern *Santa Clara, United States*

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa August 2019 - December 2019
Applied Scientist Intern *Aachen, Germany*

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

SAP AI Research May 2018 - February 2019
Research Intern / Master's Thesis *Berlin, Germany*

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

EDUCATION

TU Kaiserslautern, Germany March 2019 - June 2022
Ph.D. in Computer Science
Building Natural Language Generation and Understanding Systems in Data-Constrained Settings; work on VQA, interpretability, and conversational AI.
Overall grade: **"Sehr Gut" 1.0** (1.0 is highest on 1.0–5.0 scale)

University of Copenhagen, Denmark January 2021 - March 2021
Visiting Researcher
Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations.

TU Kaiserslautern, Germany October 2016 - November 2018
M.Sc. in Computer Science — Minor in Economics
Overall grade: **"Sehr Gut" 1.5**

Kyushu University, Japan November 2017 - February 2018
Semester Abroad
Explainable AI to analyze behavior of deep CNN architectures for image recognition.

Guru Nanak Dev Engineering College, India August 2012 - July 2016
B.Tech in Computer Science & Engineering
Overall grade: **First Division with Distinction**

SELECTED PUBLICATIONS

- **Jolly, S.**, Zhang, Z. X., Dengel, A., & Mou, L. (2022). Search and learn: Improving semantic coverage for data-to-text generation. *AAAI*.
- **Jolly, S.**, Pezzelle, S., & Nabi, M. (2021). EaSe: A diagnostic tool for VQA based on answer diversity. *NAACL-HLT*.
- **Jolly, S.**, & Kapoor, S. (2020). Can pre-training help VQA with lexical variations? *Findings of EMNLP*.
- **Jolly, S.**, Falke, T., Tirkaz, C., & Sorokin, D. (2020). Data-efficient paraphrase generation to bootstrap intent classification and slot labeling. *COLING (Industry Track)*.
- **Jolly, S.**, Iwana, B. K., Kuroki, R., & Uchida, S. (2018). How do convolutional neural networks learn design? *ICPR*. (*Best Student Paper*)

AWARDS AND HONORS

- Received **AAAI-22 Scholarship** by Hitachi to attend AAAI 2022 (2022).
- Awarded **AI Newcomer 2021 Award** (German Informatics Society and BMBF, Germany) (2021).
- Received **EU-Cost STSM Grant** to work on Multi3generation project at CopeNLU (2020).
- **Best Student Paper Award** at ICPR (2018).